Supplementary Material

Behavioural Signatures of a Language-Model Agent
in Autonomous Vision-Architecture Research

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This document accompanies the main paper. It records the implementation stack and the human-authored instruction documents that bounded the agent's behaviour (Appendix A), reproduces the complete per-hypothesis record in a single tabular projection (Appendix B), and collects additional analyses together with the tables and figures they rely on (Appendix C). The remaining supplementary artefacts (the research brief and workflow template, the full research log, the per-hypothesis CSV, the figure-generation scripts, the per-phase champion configurations, and the Slurm submission templates) are provided in the public repository `https://github.com/aonsafdar/AutoMLR` and are indexed by its `README`.

A Implementation Details

This appendix records the stack used in the case study and the human-authored documents that bounded the agent's behaviour, at a level of detail the main text omits. We do not claim that any single component is essential; we list them so that other groups can either replicate or substitute. These supplementary materials (the research log, the per-hypothesis CSV used to generate every figure in the paper, and the figure-generation scripts) are provided alongside the submission; the training code and tracking-service exports will be released under a permanent DOI upon publication.

A.1 Agent and Tooling

The case-study run used Anthropic's Claude (Opus 4 series) accessed through the Claude Code command-line interface, so that all tool use is achieved through ordinary shell and file-system primitives (file read, file edit, shell command, search). Any comparable CLI-capable frontier LLM should be a drop-in replacement. The specific model identifier and version tags at the time of each phase are recorded in the accompanying materials. No fine-tuning, RAG layer, or domain-specific scaffold was used; the agent's only persistent memory across sessions is the research log described below.

Tool access was managed through the Model Context Protocol (MCP). Three external services were exposed through MCP from Phase 1b onward: a preprint search and full-text service for arXiv, a documentation-question-answering service for GitHub repositories, and a model-and-dataset hub. Each is presented to the agent through a uniform tool interface so that literature retrieval, repository inspection, and dataset enumeration follow the same call pattern.

A.2 Compute Infrastructure

Experiments ran on a shared HPC allocation managed by Slurm, with a mix of accelerator generations. Small-scale phases used a single GPU per job; the large-scale phase used two GPUs per job (capped by the per-user allocation limit at our site). The agent was explicitly prohibited from running training on interactive login hosts; this constraint was enforced by an instruction in the research brief and, after one observed violation in early Phase 1, by a wrapper script that rejected training-command invocations outside of a Slurm job context. Aggregate compute is reported in the main paper.

A typical hypothesis cycle consisted of: literature query (a few seconds, optional), code or configuration edit (under a minute of agent wall-clock), Slurm submission (seconds), queue wait (highly variable, often the dominant component on shared infrastructure), training (minutes to days depending on phase), and result interpretation and log entry (seconds of agent time). The end-of-turn-after-submission discipline (described in the main paper) ensured that the agent did not consume tokens while the job was queued or training.

A.3 Source, Experiment, and Memory Management

A single Git repository, with one branch per phase and one commit per accepted hypothesis. Rejected hypotheses were reverted from the working tree but remain reconstructible from the research log. Commit messages followed a fixed template (`phase: short title (hypothesis ID)`), giving a one-line audit trail directly inside source control.

Experiment tracking used Weights & Biases, with one project per phase. Every Slurm job logged per-epoch training loss, validation top-1 and top-5, parameter count, FLOPs, and a fixed set of system statistics. The agent was expected to reference training curves (not only final metrics) when interpreting ambiguous results; we observed it doing so on several occasions where a final number was within noise but a divergent training curve disqualified the run.

Persistent memory took the form of a single Markdown research log containing one appended entry per hypothesis in the fixed template described in Section A.6. The agent was required to read this file in full at the start of every session (after an extended pause or context reset). By Phase 2 the log exceeded the agent's effective attention window, at which point a compact agent-authored *empirical-laws synthesis* document was introduced (regenerated periodically as a forced exercise in summarisation) and read once per session in addition to the log.

A.4 Persistence and Control Hand-off

The agent process was kept alive across user disconnects in a persistent terminal session. To return control to the agent on job completion without requiring the agent to poll the scheduler (which would have consumed tokens and triggered the failure mode of mis-classifying queued jobs as running), a short shell utility watched for the appearance of the per-job log file's termination marker and re-prompted the agent in the same terminal session. We treat this as an implementation detail rather than a methodological contribution; any equivalent control-hand-off mechanism (a daemon, a cron entry, a job-completion hook) would serve the same purpose.

A.5 Human-Authored Instruction Documents

Three human-authored documents bounded the agent's behaviour. Their exact wording (including revisions over the programme) is included in the supplementary materials. We here summarise their structure.

The research brief. The brief is two-layered: a fuller research-programme document (`RESEARCH_PROGRAM.md` in the supplementary) and a shorter agent-facing brief (`RESEARCH_BRIEF.md`) that points to it. Together they specify: the scientific question (here, designing a channel-attention vision transformer under a pure channel-attention constraint, with no spatial self-attention); the design constraints (no spatial self-attention at any stage, parameter and FLOP budget per phase, dataset for each phase); the success criterion (validation top-1 on a fixed held-out set); and a list of *illustrative* directions intended to seed the agent's literature exploration without prescribing the answer (see Section A.7 below). The brief was rewritten between phases to reflect transitions (recorded among the human interventions in the main paper); the rewritten version was always presented as the agent's authoritative single source of truth, and the previous version was archived.

The workflow template. A separate document specifying the per-hypothesis loop (described in the main paper), the expected structure of each log entry, and explicit prohibitions: no scheduler polling, no training on the interactive login host, one experiment in flight at a time, no architectural changes without a corresponding source-control commit, and a budget check (parameter count and estimated FLOPs) before any submission. This document was updated only to add reliability constraints in response to observed autonomy failures (Section A.8); the scientific content lived in the research brief.

The literature-grounding requirement. A short rule, added at the Phase 1/1b boundary: every hypothesis must either cite a concrete prior result (a paper, repository, or model card) or explicitly declare itself an internal extrapolation. This rule is short but had a visible effect on the agent's behaviour: it shifted hypothesis selection from configuration tweaks toward code-level mechanism changes. We attribute the effect to the rule forcing the agent to ground its proposal in a specific external reference rather than in pattern completion over its recent context.

A.6 Per-Hypothesis Log Template

Each entry in the research log follows a fixed seven-field template:

1. **Identifier and title.** A short symbolic identifier (`H1`, `P2-H7`, `H3-12`) and a one-line title naming the single variable changed.
2. **Motivation.** Two or three sentences stating the hypothesis to be tested and why it is worth testing now.
3. **Literature basis.** A specific citation (paper, repository, or model card) or an explicit declaration that the hypothesis is an internal extrapolation. The literature-grounding rule (Section A.5) requires this field to be non-empty from Phase 1b onward.
4. **Change description.** The exact diff applied: which configuration values, which code paths. This field is the source of the CFG/COD change-type classification used throughout the paper.
5. **Budget.** Parameter count, FLOP estimate, and expected wall-clock; all three are recomputed by a budget-check utility before submission.
6. **Result.** Final validation metric, Δ vs. current champion, and a short interpretive sentence written by the agent after the run.
7. **Status.** One of `pushed`, `discarded`, `pending`, with the corresponding Git SHA if pushed.

The full log is provided as a Markdown file in the supplementary materials. A tabular projection of all hypotheses appears in Appendix B.

A.7 Illustrative Directions in the Phase 3 Brief

The Phase 3 brief included a non-exhaustive list of five illustrative structural directions, intended to seed (not constrain) the agent's hypothesis generation: funnelling architectures with extra extreme-spatial-compression stages, factorised or low-rank channel attention, sparse top-k channel attention, hierarchical local-plus-global channel attention, and linear or state-space channel attention. The brief explicitly framed this as a *menu*, instructed the agent to synthesise novel ideas from the literature, and required a per-hypothesis self-audit of which novelty category the proposal fell into. We treat the menu as a documented direction-shaping intervention (recorded among the human interventions in the main paper), because the agent's Phase 3 hypothesis set ended up including adaptations of all five.

A.8 Autonomy Failure Modes Encountered

Several classes of autonomy failure appeared during the programme and were each fixed once. We list them because they are likely to recur in any similar setup.

1. **Premature turn termination.** In early Phase 1 the agent would sometimes declare a turn finished after merely *submitting* a Slurm job, without waiting for or interpreting the result. Fixed by an explicit "end your turn after submission and wait for an external signal" instruction (see the main paper) and the control-hand-off mechanism of Section A.4.
2. **Scheduler mis-reading.** The agent sometimes classified queued jobs as running, leading to spurious result interpretations. Fixed by routing queue-status queries through a small wrapper that returned only `queued` / `running` / `finished`, eliminating the noisy free-text scheduler output.
3. **Login-host training.** Twice in early Phase 1 the agent attempted to start a training process on the interactive login host. Fixed by a wrapper script that rejected training-command invocations outside of a Slurm job context.
4. **Distributed-data-parallel unused-parameter crash.** The agent's H3-18 hypothesis (a Synthesizer-style operator that bypasses the key projection) crashed at iteration 0 because DDP detected an unused parameter. The agent diagnosed the unused projection itself, applied a one-line fix that preserves the parameter's gradient flow without changing the numerics, and resubmitted, without external intervention.
5. **Context-window saturation.** By the middle of Phase 2 (over a hundred log entries) the research log exceeded the agent's effective attention window. Fixed by introducing the compact agent-authored synthesis document described in Section A.3.
6. **Token-rate-limit interruptions.** Long sessions occasionally hit the LLM provider's token-rate limit. Fixed by aggressively pruning what the agent was asked to re-read each turn (notes on this are in the research brief), and by tolerating short stalls without retrying through the loop.

Each failure is an instance of the same general pattern: a discipline that is implicit in a human researcher's habits (do not call a job "done" before its results are read, do not run training on a login node, summarise prior work rather than re-read it from scratch each turn) needed to be made

explicit in the agent's instructions or enforced mechanically by the loop. We expect any future autonomous-research system to need analogous discipline-encoding work; the specific list above is intended as a starting checklist.

A.9 Approximate Costs

Wall-clock: approximately ten weeks across the three phases (as summarised in the main paper), of which roughly 90% was either queue wait or training and the remaining 10% was agent activity. GPU-hours: of order 2.4 k aggregated across all phases, dominated by the large-scale ImageNet validation runs; the median substantive run was short ($\approx$ 2.5 h). LLM-token consumption: of order tens of millions of tokens across all phases (we did not log this systematically; the figure is a budget-based estimate). Human-researcher involvement: bounded to the interventions listed in the main paper, plus approximately one to two hours per day of passive oversight during active phases (monitoring for autonomy failures and unblocking when they occurred).

We make no claim that this loop is cheaper than a human researcher; our claim is that the *research work* was executed by the agent, with the human role confined to the recorded interventions, and that the resulting behavioural record is worth studying on its own.

A.10 Code and Data Availability

The supplementary materials accompanying this submission contain:

1. the two-layered research brief (`RESEARCH_PROGRAM.md` and `RESEARCH_BRIEF.md`), the workflow template (`AGENT_INSTRUCTIONS.md`), and the tooling-configuration document (`TOOLING.md`), all in their final form;
2. the research log (Markdown), with one entry per hypothesis in the template of Section A.6;
3. the agent-authored empirical-laws synthesis document referenced in Section A.3, and the agent-authored architecture-evolution summary referenced in the main paper;
4. the per-hypothesis CSV used to generate every figure and the longtable in Appendix B;
5. the figure-generation scripts (Python, $<$ 500 lines total);
6. the per-phase Slurm submission script templates and the control-hand-off watcher script (with site-specific identifiers stripped);
7. the architectural specification of the champion at the end of each phase, as YAML.

The full training code (the model implementation, training driver, and per-run configurations of failed hypotheses) is not included in this submission package and will be released under a permanent identifier upon publication. Re-running the architecture on the released training code reproduces the champion accuracy at each phase to within run-to-run variance; we did not budget for multi-seed replications of every accepted hypothesis, and this is one of the limitations discussed in the main paper.

B Per-Hypothesis Record

Table 1 reproduces the full per-hypothesis record in a single tabular projection: one row per submitted hypothesis (and one row per dataset baseline), across all four phase markers (Phase 1, Phase 1b, Phase 2, Phase 3). The table is sorted chronologically within each phase. Identifiers prefixed `H` are CIFAR-10 hypotheses (Phase 1 and Phase 1b), identifiers prefixed `P2-H` are CIFAR-100 hypotheses (Phase 2), and identifiers prefixed `H3-` are ImageNet-1K hypotheses (Phase 3). The *description* column is the agent's own one-line summary of the change, truncated to a fixed width for compactness; the full description, motivation, literature basis, change diff, budget, and outcome interpretation are in the supplementary research log (Section A.10).

Table 1: Full per-hypothesis record extracted from the research log. *Description* is truncated to the first 58 characters of the agent's title; the full motivation, literature citation, change description, and outcome interpretation are in the supplementary log. *Type* distinguishes hypotheses whose only substantive change was a configuration value (CFG) from those that introduced or modified code (COD); all Phase 3 hypotheses were structural and are labelled COD. *Acc* is the validation top-1 (%) at the dataset and protocol of that phase. Δ is the change vs. the running champion at the time the hypothesis was submitted, with the dataset baseline treated as the initial champion for each chain.

ID	Ph.	Description	Type	Acc	Δ	Status
Baseline-C10	1	Dataset baseline	–	69.67	–	
H1	1	Add CA MLP (CA_MLP_RATIO = [4.0, 4.0, 1.0]	cfg	91.08	+21.41	pushed
H2	1	Stage-2-only CA MLP (CA_MLP_RATIO = [0, 0, 1.0]	cfg	89.97	-1.11	discarded
H3	1	Multi-head channel attention (NUM_HEADS = [1, 2, 4]	cfg	89.95	-1.13	discarded
H4	1	Deeper stage-2 backbone (DEPTH = [1, 2, 11]	cfg	91.13	+0.05	pushed
H5	1	Larger early-stage CA MLP (CA_MLP_RATIO = [8.0, 8.0, 1.…	cfg	91.71	+0.58	pushed
H6	1	DW shortcut backbone (DW_SHORTCUT_BACKBONE = true	cfg	92.52	+0.81	pushed
H7	1	Remove stochastic depth (DROP_PATH_RATE = [0,0,0]	cfg	92.98	+0.46	pushed
H8	1	Shorter warmup (WARMUP_EPOCHS = 5	cfg	93.00	+0.02	pushed
H9	1	Higher learning rate (BASE_LR = 1e-3	cfg	93.77	+0.77	pushed
H10	1	Reduce weight decay (WEIGHT_DECAY = 0.01	cfg	93.36	-0.41	discarded
H11	1	Push LR further (BASE_LR = 2e-3	cfg	93.96	+0.19	pushed
H12	1	Push LR to 4e-3 (BASE_LR = 4e-3	cfg	93.41	-0.55	discarded
H13	1	4-stage architecture (NUM_STAGES=4, DIM_EMBED=[64,128,2…	cfg	94.11	+0.15	pushed
H14	1	4-stage with stage-3 stride=1 (14×14 spatial, PATCH_STRI…	cfg	93.98	-0.13	discarded
H15	1	Reallocate depth: 1 stage-1 block → 1 stage-3 block (DEPT…	cfg	93.79	-0.32	discarded
H16	1	Re-tune LR for 4-stage model (BASE_LR = 3e-3	cfg	93.84	-0.27	discarded
H17	1	Larger QKV kernel in stages 0-2 (KERNEL_QKV=[5,5,5,3]	cfg	93.39	-0.72	discarded
H18	1	dw_sep patch embedding at all 4 stages (PATCH_EMBED_ME…	cfg	93.47	-0.64	discarded
H19	1	Deeper stage 0 (DEPTH=[2,2,3,7]	cfg	93.55	-0.56	discarded
H20	1	DEPTH=[1,2,3,8] + CA_MLP_RATIO=[8,8,3.3,1] (stage-3 dep…	cfg	93.68	-0.43	discarded
H21	1	Disable Mixup+CutMix (AUG.MIXUP=0, AUG.CUTMIX=0	cfg	94.79	+0.68	pushed
H22	1	Disable all DeiT augmentation (no RandAugment, no RandomE…	cfg	94.24	-0.55	discarded

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ID	Ph.	Description	Type	Acc	Δ	Status
H23	1	Disable label smoothing (MODEL.LABEL_SMOOTHING=0.0	cfg	94.92	+0.13	pushed
H24	1	Reduce RandomErase probability (REPROB=0.1	cfg	94.71	-0.21	discarded
H25	1	Mild drop path in deepest stage (DROP_PATH_RATE=[0,0,0,…	cfg	94.96	+0.04	pushed
H26	1	Redistribute depth	cfg	94.84	-0.12	discarded
H27	1	Longer warmup (WARMUP_EPOCHS=10	cfg	95.06	+0.10	pushed
H28	1	Even longer warmup (WARMUP_EPOCHS=15	cfg	94.99	-0.07	discarded
H29	1	Progressive drop path [0, 0.01, 0.02, 0.05]	cfg	95.09	+0.03	pushed
H30	1	Layer Scale (CaiT-style, init=1e-4	cod	95.10	+0.01	pushed
H31	1	Attention dropout in stage 3 (ATTN_DROP_RATE=[0,0,0,0.1]	cfg	94.98	-0.12	discarded
H32	1	DW shortcut placed between attention and MLP (DW_SHORTCU…	cfg	95.09	-0.01	discarded
H33	1	SiLU (Swish) activation instead of GELU	cfg	94.91	-0.19	discarded
H34	1	Higher weight decay (WEIGHT_DECAY=0.1	cfg	94.99	-0.11	discarded
H35	1	Lighter RandAugment (rand-m7	cfg	95.03	-0.07	discarded
H36	1	More depth at stage 0 (DEPTH=[2,2,3,7]	cfg	94.79	-0.31	discarded
H37	1	Light Mixup (MIXUP=0.2	cfg	94.83	-0.27	discarded
H38	1	All-dw_sep patch embedding (['dw_sep','dw_sep','dw_se…	cfg	94.80	-0.30	discarded
H39	1	Deep pure CA	cfg	94.03	-1.07	discarded
H40	1	AdamW β_2 =0.98	cfg	94.74	-0.36	discarded
H41	1	Layer Scale init=1e-5	cfg	94.98	-0.12	discarded
H42	1	QK-Normalize (L2-normalize Q and K before dot product	cod	94.22	-0.88	discarded
H43	1b	Learnable Channel Pair Bias (CPB	cod	94.97	-0.13	discarded
H46	1b	Auxiliary deep supervision at stage 2 (intermediate class…	cod	95.70	+0.60	pushed
H47	1b	Per-stage learned attention temperature (log-parameterize…	cod	95.82	+0.12	pushed
H48	1b	Gated attention output (scalar sigmoid gate, channel mode	cod	95.75	-0.07	discarded
H49	1b	Larger DW shortcuts (5×5 kernels, uniform across all stag…	cfg	95.71	-0.11	discarded
H50	1b	Dual auxiliary deep supervision at stages 1 and 2	cod	95.85	+0.03	pushed
H51	1b	Convolutional Position Encoding (CPE) at each stage	cod	95.95	+0.10	pushed
H52	1b	Larger CPE kernel (5×5	cod	96.11	+0.16	pushed
H53	1b	CPE with 7×7 kernel	cod	95.85	-0.26	discarded
H54	1b	SwiGLU MLP (replace GELU with SwiGLU gate in CA MLP blocks	cod	96.23	+0.12	pushed
H55	1b	Per-stage CPE kernel sizes [3, 3, 5, 5]	cod	96.24	+0.01	discarded
H56	1b	DW shortcut kernel verified [3,3,3,3] on H54 base	cfg	96.48	+0.25	pushed
H57	1b	RMSNorm on H56 clean base	cod	96.25	-0.23	discarded
H58	1b	Multi-head channel attention NUM_HEADS [1,2,4,1]	cfg	96.12	-0.36	discarded
H59	1b	Stronger auxiliary loss weights	cod	96.42	-0.06	discarded
H60	1b	Wider stage 2 MLP (ratio 4→6), fewer blocks (3→2	cfg	96.38	-0.10	discarded
H62	1b	Depth redistribution [1,3,3,6]	cfg	96.37	-0.11	discarded
H63	1b	LayerScale 1e-3 (10x larger initial scale	cod	96.20	-0.28	discarded
H64	1b	Triple auxiliary supervision	cod	96.59	+0.11	pushed
H65	1b	Stronger aux weights	cod	96.48	-0.11	discarded
H67	1b	Intra-stage-3 mid-block auxiliary head (after block 4 of 7	cod	96.37	-0.22	discarded
Baseline-C100	2	Dataset baseline	–	81.34	–	

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ID	Ph.	Description	Type	Acc	Δ	Status
P2-H2	2	Stage-3 depth-width tradeoff	cfg	84.11	+2.77	discarded
P2-H3	2	Group channel attention (G=4) at stage 3 + deeper network	cod	81.31	-0.03	discarded
P2-H4	2	Cross-stage channel feature fusion (DuoFormer-inspired	cod	81.72	+0.38	pushed
P2-H5	2	No stochastic depth at 100ep (DROP_PATH=[0,0,0,0]	cfg	81.12	-0.60	discarded
P2-H6	2	Label smoothing LS=0.1 at 100ep	cfg	82.02	+0.30	pushed
P2-H7	2	Mild Mixup=0.4 at 100ep	cfg	82.79	+0.77	pushed
P2-H8	2	Differential channel attention	cod	81.88	-0.91	discarded
P2-H9	2	Intra-stage-3 mid-block auxiliary supervision	cod	82.80	+0.01	pushed
P2-H10	2	Periodic CPE re-injection within stage 3 (interval=4	cod	82.88	+0.08	pushed
P2-H11	2	Larger Q/K/V projection kernel at stage 3 (3×3 → 5×5	cfg	83.16	+0.28	pushed
P2-H12	2	Full-field Q/K/V projection kernel at stage 3 (5×5 → 7×7	cfg	83.11	-0.05	discarded
P2-H13	2	5×5 Q/K/V projection kernel at stage 2 as well	cfg	82.82	-0.34	discarded
P2-H14	2	5×5 DW shortcut kernel at stage 3	cfg	83.17	+0.01	pushed
P2-H15	2	Wider stage-3 embedding (576→640) with fewer depth (12→10	cfg	83.18	+0.01	pushed
P2-H16	2	Cross-stage spatial input fusion (stage-2 features → stag…	cod	82.72	-0.46	discarded
P2-H17	2	Grouped channel attention at stage 3 (G=4 groups of 160 c…	cod	82.98	-0.20	discarded
P2-H18	2	Sigmoid channel attention at stage 3	cod	82.97	-0.21	discarded
P2-H19	2	Value residual for stage-3 channel attention	cod	83.11	-0.07	discarded
P2-H20	2	Dual-scale channel attention (GAP branch	cod	83.16	-0.02	discarded
P2-H21	2	Learned channel-pair attention prior (CHANNEL_PAIR_BIAS	cod	82.89	-0.29	discarded
P2-H22	2	All-stage cross-stage fusion (add stage 0	cod	83.30	+0.12	pushed
P2-H23	2	Second-order statistics pooling in cross-stage fusion	cod	83.30	+0.00	discarded
P2-H24	2	Attentive channel gating (output-side gate on stage-3 cha…	cod	83.13	-0.17	discarded
P2-H25	2	Decoupled V projection	cod	82.82	-0.48	discarded
P2-H26	2	Per-stage periodic CPE re-injection	cod	83.37	+0.07	pushed
P2-H27	2	Stage-3 CPE interval 4→3 (denser spatial refresh in deepe…	cod	82.74	-0.63	discarded
P2-H28	2	5×5 Q/K at stage-2 (extend stage-3 winning kernel size to…	cfg	82.80	-0.57	discarded
P2-H29	2	Register tokens for stage-3 channel attention (K=4	cod	82.70	-0.67	discarded
P2-H30	2	Stage-1 periodic CPE re-injection (interval=1	cod	82.51	-0.86	discarded
P2-H31	2	7×7 CPE kernel	cod	83.03	-0.34	discarded
P2-H32	2	Depth redistribution DEPTH=[1,2,5,10]→[1,2,6,9]	cfg	83.17	-0.20	discarded
Baseline-IN1K	3	Dataset baseline	–	77.65	–	
H3-1	3	Sparse top-k channel attention at stage-3 (k=64 of C=640	cod	77.48	-0.17	discarded
H3-2	3	Dual-resolution channel attention at stage-3 (G=32 super-…	cod	77.50	-0.15	discarded
H3-3	3	Cross-stage channel attention bridge (stage-2 → stage-3	cod	77.55	-0.10	discarded
H3-4	3	Agent attention at stage-3 (factorised low-rank CA via M=…	cod	77.38	-0.27	discarded
H3-5	3	Hierarchical local + global CA at stage-2 (within-group f…	cod	77.45	-0.20	discarded

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ID	Ph.	Description	Type	Acc	Δ	Status
H3-6	3	Funneling tail-block at 3×3 spatial pool (purely additive…	cod	77.59	-0.06	discarded
H3-8	3	Linear-attention funneling tail at full stage-3 spatial (…	cod	62.93	-14.72	discarded
H3-9	3	Linformer-style projected channel attention funnel + shar…	cod	77.79	+0.14	pushed
H3-10	3	Stacked Linformer-style projected channel attention funne…	cod	77.75	-0.04	discarded
H3-11	3	Parallel hybrid funnel	cod	77.58	-0.21	discarded
H3-12	3	Linformer applied to ALL 10 stage-3 in-trunk CA blocks (L…	cod	77.91	+0.12	pushed
H3-13	3	Push Linformer wider	cod	77.65	-0.26	discarded
H3-14	3	Per-block Linformer rank schedule within stage-3	cod	77.67	-0.24	discarded
H3-15	3	Sparse top-k WITHIN the Linformer rank-128 supertoken bas…	cod	77.54	-0.37	discarded
H3-16	3	Gated parallel FULL-RANK CA pathway alongside Linformer a…	cod	77.78	-0.13	discarded
H3-17	3	Sparsemax replacement for softmax inside stage-3 Linforme…	cod	77.74	-0.17	discarded
H3-18	3	Synthesizer-Dense channel attention at stage-3 (REPLACES …	cod	77.39	-0.52	discarded

C Additional Tables, Figures and Analyses

This appendix expands three analyses that Part 2 of the main paper summarises: the attribution of the CIFAR-10 gain, the qualitative decomposition of the agent's behavioural biases into workflow-induced and inherent components, and the cross-scale reversals of training-recipe decisions. For convenience it also reproduces, from the main paper, the phase summary (Table 2), the complete record of human interventions (Table 3), the Phase-1/1b tool-access transition (Table 4) and the process figure (Figure 1).

C.1 Tables and figure reproduced from the main paper

Table 2: The phases of the programme, separated by human-declared transitions that expanded the tool surface (Phase 1→1b) or changed the problem scale (Phases 2, 3). Identifiers are the agent's own labels (some numbers skipped; counts are of submitted hypotheses). "Accepted" hypotheses improved on the running champion and were committed.

Phase	Identifiers	Dataset	Budget	Tool surface	Submitted	Accepted
1	H1–H42	CIFAR-10	5.8M	config editing, tracking, source control; code edits not yet authorised	42	15 (36%)
1b	H43–H67	CIFAR-10	5.8M	+ literature access; code edits authorised; grounding rule	21	8 (38%)
2	P2-H1–P2-H32	CIFAR-100	22M	full toolkit; larger budget	31	10 (32%)
3	H3-1–H3-18	ImageNet-1K	22–27M	+ multi-GPU training; 100-ep screening, 300-ep validation	17	2 (12%)
Total					111	35 (32%)

Table 3: Complete record of human interventions. All are strategic; every tactical decision (hypothesis selection, configuration editing, code writing, result interpretation) was made by the agent. Rows marked † shaped the direction of search and are discussed in the text.

When	Type	What changed
Phase 1 start	Research brief issued	Scientific question, design constraints, parameter and compute budgets, workflow template.
During Phase 1	Reliability repair	Agent told to end its turn after submission rather than poll the scheduler.
Phase 1 / 1b boundary	Tool-surface expansion	Literature access enabled; literature grounding required; code-level changes authorised.
Phase 1b / 2 boundary	Scale transition†	Budget raised to 22M; dataset changed to CIFAR-100; note to "re-tune the training recipe at the new scale".
During Phase 2	Direction note†	Plateau flagged; brief amended to prioritise structural innovation over configuration tuning.
During Phase 2	Memory discipline	Full research-log read required at session start; compact agent-authored synthesis document introduced.
Phase 2 / 3 boundary	Novelty discipline†	"Bold structural hypothesis" requirement, per-entry novelty self-audit, and a menu of five illustrative directions (all five later adapted by the agent).
Phase 3 start	Benchmark transition	Primary benchmark switched to ImageNet-1K with 100-epoch screening and 300-epoch validation.
End of Phase 3	Programme closure	Stopped after saturation around the low-rank attention champion; no new direction prescribed.

Table 4: The Phase-1 → Phase-1b transition (literature access granted, code-level changes authorised, grounding rule added). The transition did not increase per-hypothesis gain magnitude but changed the *kind* of hypothesis proposed, ending the 29-hypothesis saturation plateau. H1 is the +21.4 pp feed-forward hypothesis.

Metric	Phase 1	Phase 1b
Code changes (frac. of hypotheses)	2/42 = 5%	16/21 = 76%
Novel mechanisms introduced	2	8
Distinct papers cited	0	24+
Avg gain / successful hyp.	+1.70 pp	+0.19 pp
Avg gain / succ. hyp., excl. H1	+0.29 pp	+0.19 pp

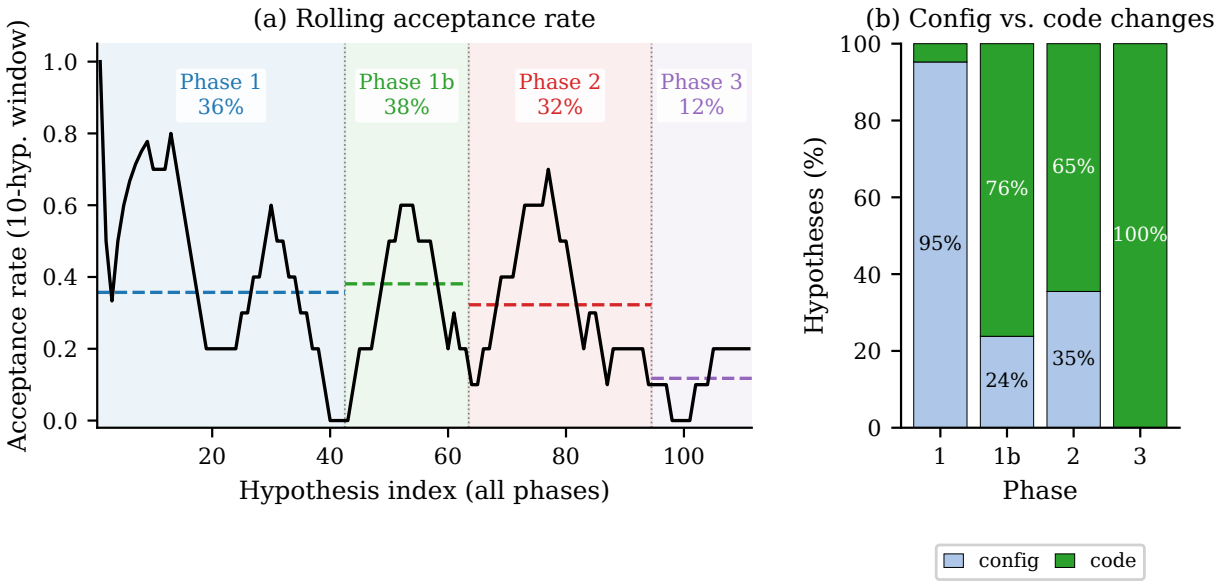


Figure 1: Process signature of the programme (reproduced from the main paper). (a) Rolling ten-hypothesis acceptance rate across all 111 hypotheses; dashed lines are per-phase means. (b) Fraction of hypotheses that modified code rather than only configuration, by phase.

C.2 Attribution of the accuracy gain

Attributing the overall CIFAR-10 improvement of 26.9 percentage points to individual accepted hypotheses yields a strongly long-tailed distribution (Figure 2). The single largest gain comes from the very first accepted hypothesis (H1), adding a feed-forward sublayer to every channel-attention block, which alone accounts for +21.4 pp, roughly four-fifths of the total. The baseline the agent inherited had no feed-forward sublayer anywhere, so H1 is best read as the agent repairing an obviously incomplete block; its magnitude says more about the starting point than about the agent's later search. Training-recipe tuning across H7–H29 (learning rate, warm-up, augmentation, drop-path) adds ≈2.6 pp; every architectural refinement thereafter is sub-point. This shape characterises the trajectory as a whole: a few early decisions dominate the outcome, while the long autonomous tail contributes steady but sharply diminishing returns. Because runs are single-seed, the smallest accepted deltas (Layer Scale +0.01, warm-up +0.02, drop-path +0.03/+0.04) cannot be distinguished from run-to-run variation; they were accepted because the commit-or-discard rule tests only for a positive sign.

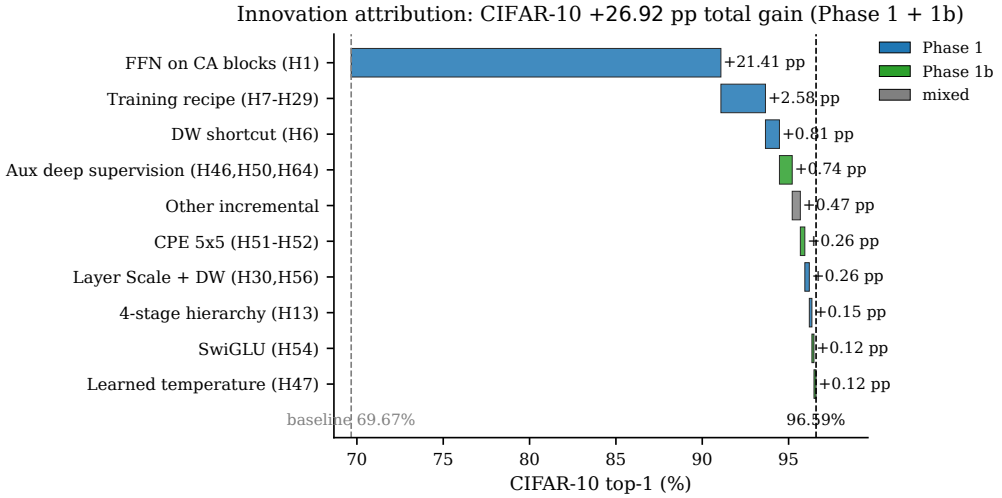


Figure 2: Attribution of the CIFAR-10 improvement (+26.9 pp) to grouped hypotheses. The first accepted hypothesis (H1, feed-forward sublayer) accounts for roughly four-fifths of the gain; every later accepted hypothesis is a modest refinement.

C.3 Workflow-induced versus inherent bias

Figure 3 summarises a qualitative decomposition of the agent's observed behavioural biases (greedy anchoring, post-failure risk aversion, and the overall drift toward incrementalism) into a component attributable to the greedy commit-or-discard workflow we imposed and a residual attributable to the agent's own tendencies. The workflow-induced component is the larger of the two and is directly addressable by a workflow change (maintaining a population of parallel champions, budgeting mandatory moonshots, supporting explicit backtracking); the residual is smaller but harder to modify. We stress that this is a qualitative decomposition, offered as a structured hypothesis for a future multi-workflow study to test, and not a controlled ablation. It is the reasoning behind the claim in Part 2 that the loop is at least as consequential as the model.

C.4 Cross-scale reversals

A subset of Phase-1 findings did not transfer to later phases, and in several cases reversed outright (Figure 4). Recipe settings tuned at the small (5.8M, CIFAR-10) scale (disabling Mixup, disabling label smoothing, and removing drop-path) were carried into the larger (22M, CIFAR-100) scale, where all three flipped sign, and recovering the lost accuracy required several hypotheses of re-tuning. The agent did not flag the scale transition itself; re-validation was triggered only by an explicit human note at the phase boundary. Because the phases also differ in dataset and epoch budget, part of each reversal may be protocol-driven rather than scale-driven; the behavioural point is unaffected, since the agent re-opened none of the decisions unprompted. The practical lesson for autonomous-research systems is that a decision validated at one scale should be treated as *locally valid* and re-checked when the regime changes; the

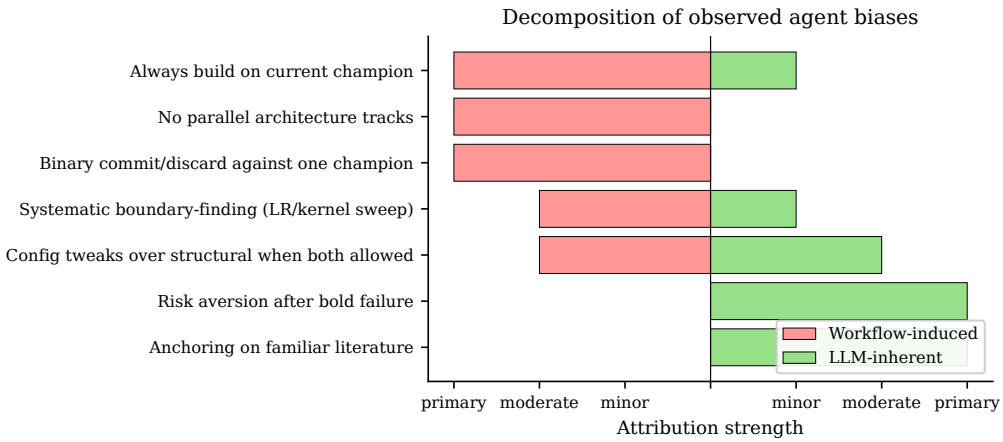


Figure 3: Diverging-bar decomposition of the observed behavioural biases. Left bars are the component attributable to the imposed greedy commit-or-discard workflow; right bars are the residual attributable to the agent's own tendencies. Bar length encodes the authors' qualitative judgement of attribution strength from the research log. This is a qualitative decomposition, not a controlled ablation.

simplest mitigation is to annotate every accepted finding with the regime in which it was validated and to re-validate the leading findings whenever that regime changes.

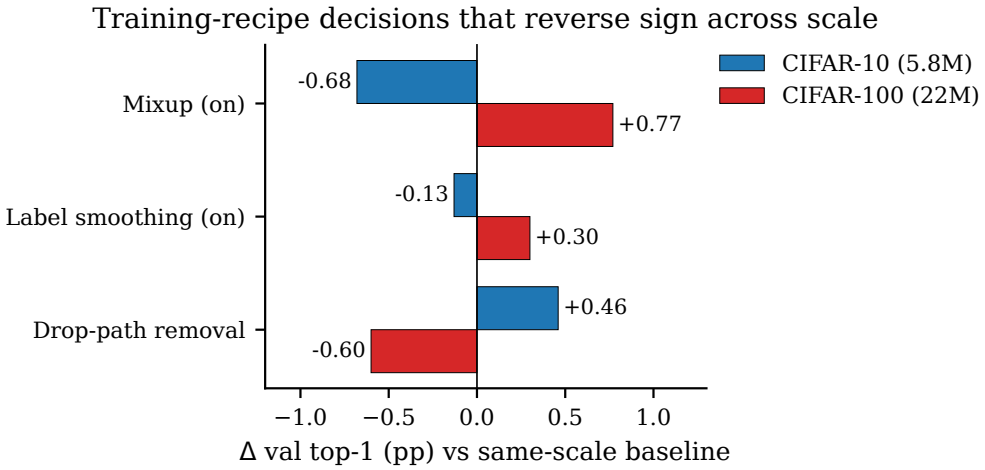


Figure 4: Cross-scale reversals. Three training-recipe decisions tuned at the small (5.8M, CIFAR-10) scale reversed sign at the larger (22M, CIFAR-100) scale: Mixup and label smoothing, mildly harmful at the small scale, became beneficial; drop-path removal, beneficial at the small scale, became harmful.